SAKASENSE · KASAKA

Soil Reading Guidebook for Farmers

What your SakaSense numbers mean, and what to do about them, for the four vegetables most grown around Tacloban City and Leyte.

Kangkong
Upland water spinach · codes 72xx

Pechay
Petsay · codes 71xx

Sweet potato
Kamote · codes 23xx

Okra
Codes 31xx

Region VIII Edition · Eastern Visayas

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How to use this guidebook

Your SakaSense device measures four things in your soil: **pH** (sourness), **Nitrogen (N)**, **Phosphorus (P)** and **Potassium (K)**. This book tells you whether each number is right for your crop, and what to add if it is not.

1. **Take a reading** the right way (page 4). A bad sample gives a wrong answer.
2. **Open your crop's chapter.** Kangkong p. 8, Pechay p. 12, Sweet potato p. 16, Okra p. 20.
3. **Find each number in the reading table.** Every number falls in one of four boxes: ✕ LOW | ! SLIGHTLY LOW | ✓ GOOD | ▲ HIGH
4. **Read the code.** The device screen and the Kasaka app show a 4-digit code, like `7210`. Look it up in the crop's *Fix-it* page or the code finder on page 27.
5. **Do the fix, then test again** 2 weeks later. Write it down in the record sheet (page 27).

Amounts in this book

All amounts are given for **one bed of 1 m × 20 m (20 m²)**, the standard bed size of the Department of Agriculture vegetable guides. The amount per hectare is shown beside it. One 50-kg bag per hectare equals **100 g per bed**. No scale? Use the spoon and cup table on page 25.

Please read

The SakaSense probe gives a **quick field estimate**, not a laboratory test. Use it to decide *what to do this week*. For a new field, or once a year, bring a soil sample to the **DA Regional Field Office 8 Regional Soils Laboratory** or ask your City/Municipal Agriculture Office about the Mobile Soil Laboratory (page 26). If a lab report and your probe disagree, follow the lab.

Always follow the label and safety instructions on fertilizer bags.

Where the numbers come from

Soil conditions in Leyte: Bureau of Soils and Water Management (BSWM) Soil Health Assessment maps, made with DA-RFO 8 and LGUs. Crop pH, planting and fertilizer practice: DA Regional Field Unit and Agricultural Training Institute production guides, and Visayas State University (VSU) research in Leyte. Full list on page 28.

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Every crop chapter has the same four pages: 1 · how to grow it, 2 · the reading table, 3 · the standard fertilizer plan, 4 · the Fix-it codes.

Taking a good soil reading

One reading from one spot can mislead you. Soil changes from one corner of a bed to the next, so always take **five readings per bed** and use the average.

1. **Pick 5 spots** in a zig-zag (W shape) across the bed. Stay away from the edges, compost piles and places where fertilizer was just put.
2. **Clear the surface.** Push away grass, mulch and stones.
3. **Make sure the soil is moist**, not dry and not flooded. If it is dry, water the spot and wait 30 minutes.
4. **Push the probe in to the root zone** (about 10–15 cm deep). The metal parts must touch the soil fully, with no air gaps.
5. **Press Measure** on the device or in the Kasaka app and keep the probe still until the numbers stop changing (up to 45 seconds).
6. **Write down pH, N, P and K.** Do the same at the other 4 spots, then add each number up and divide by 5.
7. **Wipe the probe clean** with clean water and a cloth. Store it dry.

When to test

- **3 weeks before planting**, so there is time to lime.
- **Mid-season**, 2 weeks after side-dressing.
- **After harvest**, to plan the next crop.

Do not test

- within **7 days** after fertilizing
- within **3–4 weeks** after liming
- in waterlogged soil, or right after heavy rain

The device may also show temperature and humidity. These are for your records only; the soil advice in this book uses pH, N, P and K.

Understanding pH, N, P and K

pH: how sour or bitter the soil is



◀ Sour (acidic) Best for most vegetables: 5.5–7.0 Bitter (alkaline) ▶

pH is the key that unlocks food in the soil. When soil is too sour, the fertilizer you apply gets **locked up** and the plant cannot use it, so you waste money. **Fix pH first, then fertilize.**

NUTRIENT	WHAT IT DOES	SIGNS IT IS LOW
N Nitrogen	Green leaves and fast growth. Leafy crops (kangkong, pechay) need the most.	Pale or yellow older leaves, slow small plants
P Phosphorus	Strong roots, flowers and pods. Very important at planting.	Weak roots, dark or purplish leaves, few flowers
K Potassium	Root size, pod quality, strong stems and resistance to disease. Kamote needs the most.	Brown, burnt leaf edges; small roots; weak stems

The four reading boxes

✕ LOW	Clearly not enough. Act now with the **full** fix dose.
! SLIGHTLY LOW	A little short. Use **half** the fix dose.
✓ GOOD	Right for the crop. Follow the standard plan only.
▲ HIGH	More than enough. **Skip** fertilizer with that nutrient; it is wasted money and can harm the crop.

N, P and K are shown in **mg/kg** (the same as ppm). These are the units on your SakaSense screen and in the Kasaka app.

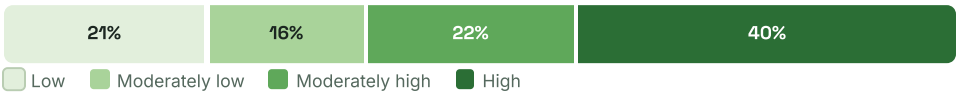
The soils of Tacloban and Leyte

The Bureau of Soils and Water Management, with DA Regional Field Office 8 and the LGUs, tested farm soils across **15,000 hectares** of key farm areas in Leyte province (Soil Health Assessment, maps published 2018). Here is what they found.

Soil pH — share of tested area



Available phosphorus (P) — BSWM rating



Extractable potassium (K) — BSWM rating



What this means for your farm

- **Most soils have a good pH** (5.6–6.8), but about 1 in 5 farms is sour enough to need lime. Around Tacloban, Palo and Santa Fe the maps show mostly good pH with some strongly acidic patches.
- **Phosphorus is the most common shortage.** Almost 4 in 10 farms are low or moderately low. Low-P patches are common between Tanauan, Tabontabon, Dagami and Tolosa.
- **Potassium is usually enough.** 6 in 10 farms rate high, so many farmers can use less potash. Test first before buying.
- The very high-pH (alkaline) soils are mostly on **limestone in western Leyte**, not around Tacloban.

Every farm is different, so these are averages, not your result. Your own SakaSense reading tells you about *your* bed. Southern Leyte farms tested by the DA RFO 8 Mobile Soil Laboratory in 2026 were also found slightly to highly acidic.

Source: BSWM, Province of Leyte Soil pH Map and Nutrient Status Maps (P, K), Key Corn Areas, 1:130,000, Soil Health Assessment Phase III with DA-RFO 8 and LGUs, 2018. K percentages recalculated from the map's hectare figures.

Which crop suits my soil?

	KANGKONG	PECHAY	SWEET POTATO	OKRA
Good pH	5.5–7.0	5.5–7.0 best 6.0–6.5	5.5–6.8	5.5–7.0 best 6.0–6.8
Copes with sour soil?	Fair	Poor	**Best**	Fair
Needs most	N	N	K	N and P
Water	A lot, every day	Every day	Little; hates standing water	When soil is dry
Rainy months	Very good	Raised beds and shelter	High ridges	Raised beds
First harvest	3–4 weeks after planting	3–4 weeks after transplanting	3–4 months (modern varieties)	Pods from 40–75 days
Codes	72xx	71xx	23xx	31xx

Quick choices

- **pH between 5.0 and 5.4 and no money for lime?** Plant **sweet potato**. It tolerates sour soil best of the four.
- **Low spot that stays wet?** Plant **kangkong**. Pechay, okra and sweet potato all suffer in waterlogged soil.
- **Need cash fast? Pechay** and **kangkong** are ready in about a month.
- **Low nitrogen and little money for fertilizer? Sweet potato** needs the least nitrogen. Too much N actually makes small roots.
- **Rotate.** Do not plant pechay in the same bed season after season; it invites club root. Follow it with sweet potato or okra.

Tacloban weather: rain falls all year, heaviest from November to January, and typhoons come often. Use raised beds (15–20 cm) and drainage canals for pechay and okra during the wet months. Sweet potato vines lie low and recover well after strong winds.

Kangkong

Ipomoea aquatica / I. reptans · upland water spinach

Codes
72xx

GOOD PH	FIRST HARVEST	SPACING
5.5 – 7.0	3–4 weeks	20 × 20 cm

Growing kangkong

1. **Soil.** Grows in most soils; best in sandy loam with good drainage and lots of organic matter.
2. **Prepare the land.** Plow and harrow 2–3 times, one week apart, 15–20 cm deep. Make beds 1 m wide × 20 m long. Mix in the basal fertilizer (page 10). Plastic mulch is optional.
3. **Plant.** Sow 2–3 seeds per hill in 4 rows per bed, 20 cm between rows and between hills. Stem cuttings (25–30 cm, 2–3 nodes in the soil) also work. Kangkong can also be grown in containers.
4. **Water** right after planting and **every day** after. Kangkong never wants to dry out.
5. **Weed** often in the first 3 weeks, until the plants cover the bed.
6. **Harvest** 3–4 weeks after planting by cutting the shoots about 5 cm above the soil, or pull whole plants. Cut plants regrow; cut again every 2–3 weeks and feed after each cut.

Pests and diseases
Caterpillars, whiteflies and aphids are the common pests. Keep the bed clean, remove weeds and damaged plants, and check under the leaves twice a week. Use resistant varieties and biological or chemical control only when needed, following the label.

Source: DA RFU VII Guide on Vegetable Production / ATI-VII Kangkong Production Guide (2020); BPI Kangkong Production Guide.

Kangkong: reading your numbers

Find each of your averaged numbers in its row. The box it falls in tells you what to do.

	✕ LOW	! SLIGHTLY LOW	✓ GOOD	▲ HIGH
pH	below 5.0 too sour	5.0 – 5.4 a bit sour	5.5 – 7.0 best 5.5–6.5	above 7.0 alkaline
N mg/kg	below 25	25 – 39	40 – 100	above 100
P mg/kg	below 8	8 – 14	15 – 40	above 40
K mg/kg	below 50	50 – 79	80 – 200	above 200

Example

Average of 5 readings: **pH 5.8 · N 30 · P 18 · K 95**

- pH 5.8 → ✓ GOOD
- N 30 → ! SLIGHTLY LOW
- P 18 and K 95 → ✓ GOOD

Code **7210** (nitrogen low). Follow the standard plan and add **half** the N fix: 50 g urea per bed (page 11).

Why these numbers?

pH range from the DA RFU VII / ATI-VII kangkong guide (5.5–6.5 best); kangkong tolerates slightly higher pH. P and K boxes follow the BSWM soil ratings used for Leyte: "high" P starts at about 15 mg/kg (Olsen) and "high" K at about 75 mg/kg. Kangkong is a leafy crop, so it needs a good supply of nitrogen, but an economic-rate study on acid soils found it responds little to potassium.

Kangkong: standard fertilizer plan

Use this plan when all four readings are ✓ GOOD. If any reading is low or high, change it with the Fix-it page.

WHEN	PER BED (1 × 20 M)	PER HECTARE
Basal at bed making, before planting	1 sack dried chicken manure **300 g complete 14-14-14** (about 1¼ cups) Mix into the top soil or cover with soil.	3 bags 14-14-14 + manure
Week 1–2 after the seedlings come up	**100 g urea 46-0-0** (½ cup) dissolved in 16 L water. Pour about 150 mL beside each hill, then water.	1 bag urea
After every cutting	**100 g urea** dissolved in 16 L water, applied the same way.	1 bag urea per cut

Tips

- Apply urea in the late afternoon and water it in. Urea left on dry soil in the sun is lost to the air.
- Never let dry urea touch the leaves; it burns them.
- Organic option: replace the urea with compost tea or fermented plant juice every week.
- Kangkong grown in very wet or flooded beds still needs feeding after each cutting.

Note on the DA leaflet: the ATI-VII leaflet lists 1 kg 14-14-14 and 4 kg urea per bed. That is more than 10 times the rate shown in research and is very costly. This book uses lower, research-based amounts. Ask your agriculture technician if you want to use the leaflet rate.

Basal method and drench (16 L, 150 mL per hill): DA RFU VII / ATI-VII. Rates set near the economic optimum of about 40 kg N and 40 kg P_2O_5 per ha found for kangkong on acid Ultisols (SANREM/IPB study), with manure supplying extra P and K.

Kangkong: Fix-it codes

Amounts are **in addition to** the standard plan. For ! SLIGHTLY LOW use **half**.

7201 All good	Follow the standard plan (page 10). Test again in 2–3 weeks.
7202 Too sour	**Lime first.** Spread 4 kg agricultural lime or dolomite per bed (2 kg if only slightly sour), mix into the top 15 cm and water. Wait 2 weeks before planting or putting urea. See page 24.
7203 Alkaline	Do not use lime or wood ash. Use **ammonium sulfate 21-0-0** instead of urea (2 g for every 1 g urea) and work in 1 extra sack of compost.
7210 N low	Add **100 g urea** per bed, in two parts one week apart, dissolved in water. Check that leaves turn darker green within a week.
7211 P low	At planting, mix **300 g ammonium phosphate 16-20-0** (or 400 g Solophos 0-18-0) into the planting rows. On a growing crop, place it in a shallow furrow beside the rows, cover and water.
7212 K low	Add **100 g muriate of potash 0-0-60** per bed, spread between rows and watered in.
7220 Several low	Add **300 g more 14-14-14** to the basal and 1 more sack of manure or compost. Test again after 2 weeks and fix whatever is still low. If pH is also low, do 7202 first.
HIGH ▲ any	N high: skip urea this round. P high: use urea + 0-0-60 instead of 14-14-14. K high: skip 0-0-60.

Pechay

Brassica rapa L. · petsay

Codes
71xx

GOOD PH	HARVEST	SPACING
5.5 – 7.0	3–4 wk after transplant	10 × 20 cm

Growing pechay

1. **Soil.** Sandy to clay loam with plenty of organic matter. Pechay grows best at pH 6.0–6.5.
2. **Prepare the land.** Plow and harrow 2–3 times, one week apart, 15–20 cm deep. Beds 1 m × 20 m. Mix in basal fertilizer (page 14).
3. **Raise seedlings** in a raised seedbed (1 m × 10 m, 15 cm high, fine sterilized soil) or in seedling trays with 2–3 seeds per hole. Cover lightly and sprinkle with water.
4. **Seedling care.** Shade from strong sun and heavy rain. Water morning or afternoon. Harden for one week before transplanting by giving more sun each day.
5. **Transplant** 2–3 weeks after sowing, in the late afternoon, 10 cm between plants and 20 cm between rows. Water right away and then daily.
6. **Harvest** 3–4 weeks after transplanting, early morning or late afternoon. Cut with a sharp knife or pull with the roots and wash.

Pests and diseases

Pests: aphids, flea beetles and diamondback moth. Diseases: powdery mildew on seedlings, bacterial soft rot, and **club root** (swollen, knotted roots; plants wilt at midday). Pull and bury sick plants. Club root is worse in sour soil, so keeping pH near 6.5 and rotating crops helps prevent it.

Source: DA RFU VII Guide on Vegetable Production / ATI-VII Pechay Production Guide (2020); DA Caraga pechay techno guide (pH 5.5–6.5).

Pechay: reading your numbers

Find each of your averaged numbers in its row.

	✕ LOW	! SLIGHTLY LOW	✓ GOOD	▲ HIGH
pH	below 5.0 too sour	5.0 – 5.4 a bit sour	5.5 – 7.0 best 6.0–6.5	above 7.0 alkaline
N mg/kg	below 30	30 – 49	50 – 110	above 110
P mg/kg	below 12	12 – 19	20 – 50	above 50
K mg/kg	below 60	60 – 99	100 – 220	above 220

Example

Average of 5 readings: **pH 5.2 · N 55 · P 10 · K 120**

- pH 5.2 → ! SLIGHTLY LOW (a bit sour)
- P 10 → ✕ LOW
- N 55 and K 120 → ✓ GOOD

pH comes first: code **7102**. Lime with 2 kg per bed, wait 2 weeks, then plant using the P fix of code **7111**.

Why these numbers?

pH: DA guides give 5.5–6.5; growers aim for 6.0–6.5, and liming to near 6.5 lowers club root. Pechay is a fast, leafy brassica that needs more nitrogen and phosphorus than kangkong. A Philippine recommended rate for pechay is about 60-40-60 kg N-P_2O_5-K_2O per hectare, so the P and K "good" boxes start higher than for kangkong.

Pechay: standard fertilizer plan

Use this plan when all four readings are ✓ GOOD.

WHEN	PER BED (1 × 20 M)	PER HECTARE
Basal at bed making	1 sack dried chicken manure **400 g complete 14-14-14** (about 1½ cups) **50 g muriate of potash 0-0-60** (3 tbsp) Mix into the soil before transplanting.	4 bags 14-14-14 ½ bag 0-0-60 + manure
1 week after transplanting	**50 g urea** (5 tbsp) dissolved in 16 L water; about 150 mL beside each plant.	½ bag urea
2 weeks after transplanting	**50 g urea** in 16 L water, same way.	½ bag urea
Seedlings	Drench trays with a weak starter solution (1 tbsp 14-14-14 dissolved in 16 L water) a few days before transplanting.	—

Tips

- Stop urea at least **7 days before harvest**. Late nitrogen makes leaves soft and lowers keeping quality.
- Water every day, but do not let water stand in the bed. In the rainy months use raised beds with canals.
- Organic option: 2 sacks of compost or vermicast in place of the 14-14-14, plus weekly compost tea.

Basal method, drench and timing: DA RFU VII / ATI-VII. Total about 51-28-43 kg N-P_2O_5-K_2O per ha from fertilizer plus manure, close to the 60-40-60 recommended rate used in Philippine pechay research. The ATI leaflet's 4 kg urea per bed is far above this rate.

Pechay: Fix-it codes

Amounts are **in addition to** the standard plan. For ! SLIGHTLY LOW use **half**.

7101 All good	Follow the standard plan (page 14). Test again after harvest.
7102 Too sour	**Lime first.** 4 kg agricultural lime or dolomite per bed (2 kg if only slightly sour), mixed into the top 15 cm and watered. Wait 2 weeks before transplanting. Liming also helps prevent club root. See page 24.
7103 Alkaline	No lime or wood ash. Use **ammonium sulfate 21-0-0** in place of urea (2 g for every 1 g urea). Add 1 extra sack of compost.
7110 N low	Add **100 g urea** per bed: 50 g at 1 week and 50 g at 2 weeks, on top of the standard doses, dissolved in water.
7111 P low	Mix **300 g ammonium phosphate 16-20-0** (or 400 g Solophos 0-18-0) into the bed before transplanting. Phosphorus barely moves in soil, so it must be near the roots.
7112 K low	Add **100 g muriate of potash 0-0-60** per bed, mixed in at basal.
7120 Several low	Add **300 g more 14-14-14** and 1 more sack of manure or compost at basal. Test again 2 weeks after transplanting and fix what is still low. If pH is also low, do 7102 first.
HIGH ▲ any	N high: skip urea this round. P high: use urea + 0-0-60 instead of 14-14-14. K high: skip 0-0-60.

Sweet potato

Ipomoea batatas (L.) Lam. · kamote

Codes
23xx

GOOD PH	HARVEST	SPACING
5.5 – 6.8	3–4 months	75 × 25 cm

Growing sweet potato

1. **Soil.** Loose, well-drained sandy loam. It tolerates sour, poor soil better than the other three crops, but **not standing water**.
2. **Planting material.** Use healthy tip cuttings 25–30 cm long from vigorous, pest-free plants. Clean planting material of improved varieties is available from **VSU PhilRootcrops** in Baybay City and DA RFO 8.
3. **Make ridges** 75 cm apart and 25–30 cm high. Ridges keep roots dry in the rainy season.
4. **Plant** one cutting per hill, 25 cm apart, with 2–3 nodes buried, slanting into the ridge.
5. **Weed and hill up** at about 2–4 weeks and again at 1 month, pulling soil back up the ridge. Cover any soil cracks to keep weevils out.
6. **Lift the vines** once a month so they do not root at the nodes.
7. **Harvest** 3–4 months after planting for modern varieties (traditional varieties take up to 6 months).

Pests and diseases

The worst pest is the **sweet potato weevil**, which bores into the roots and makes them bitter. Prevent it by using clean cuttings, hilling up to close soil cracks, harvesting on time, and removing old roots and vines after harvest. Rotate with a non-*Ipomoea* crop; kangkong is a relative of kamote, so do not rotate kamote with kangkong.

Source: VSU research in Leyte (Asio et al., Tropentag 2018: 0.75 × 0.25 m spacing, ridges); DA RFO 02 Sweet Potato Production Guide (2012); PhilRootcrops, VSU.

Sweet potato: reading your numbers

Find each of your averaged numbers in its row. Note the nitrogen row: for kamote, **too much nitrogen is a problem**.

	✕ LOW	! SLIGHTLY LOW	✓ GOOD	▲ HIGH
pH	below 4.5 too sour	4.5 – 5.4 a bit sour	5.5 – 6.8 best 5.6–6.5	above 6.8 alkaline
N mg/kg	below 15	15 – 24	25 – 60	above 60 all vines, few roots
P mg/kg	below 8	8 – 14	15 – 40	above 40
K mg/kg	below 75	75 – 119	120 – 250	above 250

Example

Average of 5 readings: **pH 5.1 · N 70 · P 16 · K 90**

- pH 5.1 → ! SLIGHTLY LOW. Kamote can grow here; liming is optional.
- N 70 → ▲ HIGH: skip all extra nitrogen.
- K 90 → ! SLIGHTLY LOW

Code 2312. Use 0-0-60 only (half the K fix, 75 g per bed) and no urea.

Why these numbers?

In a VSU trial in Leyte (sandy loam, pH 5.8, P 19 mg/kg, high K), root yield was highest at 80–160 kg N per ha and dropped sharply above that, because extra nitrogen grows vines instead of roots. The recommended rate there was 40-40-60 kg N-P_2O_5-K_2O per ha; phosphorus above 40 kg gave no gain. Kamote takes up more potassium than nitrogen, so its "good" K box is the highest of the four crops.

Sweet potato: standard fertilizer plan

Use this plan when all four readings are ✓ GOOD. It supplies about **40-40-60** kg N-P_2O_5-K_2O per hectare, the rate used in VSU trials in Leyte.

WHEN	PER 20 M² (ONE 1 × 20 M AREA)	PER HECTARE
At planting	**200 g complete 14-14-14** (¾ cup) **50 g muriate of potash 0-0-60** (3 tbsp) Put in a furrow along the ridge, cover with 3–5 cm soil, then plant. Do not let fertilizer touch the cuttings.	2 bags 14-14-14 ½ bag 0-0-60
About 1 month after planting at hilling-up	**200 g 14-14-14** + **50 g 0-0-60**, placed beside the hills and covered when you hill up.	2 bags 14-14-14 ½ bag 0-0-60

Splitting the dose in two follows DA guidance: when the rate is more than 45 kg per ha, apply half at planting and half one month later.

Tips

- **No urea** in the standard plan. Kamote rarely needs it, and too much makes long vines and small roots.
- Do not fertilize after 2 months. Late fertilizer goes to the leaves, not the roots.
- Compost or rice-hull ash worked into the ridge adds potassium and loosens heavy soil.
- Where the soil is already rich (good N, P and K), many Leyte farmers grow kamote with manure only.

Rates: Asio, Tamayo & dela Cruz (VSU/CLSU), Response of a Traditional Sweetpotato Variety to Fertilization in Leyte, Tropentag 2018. Split application: DA RFO 02 Sweet Potato Production Guide.

Sweet potato: Fix-it codes

Amounts are **in addition to** the standard plan, per 20 m². For ! SLIGHTLY LOW use **half**.

2301 All good	Follow the standard plan (page 18).
2302 Too sour	Below pH 4.5: spread **2 kg agricultural lime or dolomite** per 20 m², mix into the soil and wait 2 weeks before planting. Between 4.5 and 5.4 liming is optional; kamote tolerates it. See page 24.
2303 Alkaline	No lime or ash. Work in 1–2 sacks of compost. If you must add N, use ammonium sulfate 21-0-0.
2310 N low	Add only **50 g urea** at hilling-up (1 month), never more. Do not repeat it.
2311 P low	Mix **200 g ammonium phosphate 16-20-0** (or 250 g Solophos 0-18-0) into the planting furrow.
2312 K low	Add **150 g muriate of potash 0-0-60**: half at planting, half at hilling-up. Potassium makes bigger, better roots.
2320 Several low	Add **200 g more 14-14-14** and 1 sack manure or compost at planting. Test again at 1 month and fix what is still low. If pH is below 4.5, do 2302 first.
HIGH ▲ any	**N high (above 60): use no 14-14-14 at all;** feed with 0-0-60 only (100 g per 20 m²). P high: use 0-0-60 alone. K high: skip 0-0-60.

Okra

Abelmoschus esculentus (L.) Moench

Codes
31xx

GOOD PH	FIRST PODS	SPACING
5.5 – 7.0	40–75 days	20 × 25 cm

Growing okra

1. **Soil.** Grows in any soil but best in sandy loam, pH 5.5–7.0 (best 6.0–6.8). Choose a high-yielding, pest-resistant variety that buyers in your market like.
2. **Prepare the land.** Plow and harrow 2–3 times, one week apart, 15–20 cm deep. Beds 1 m × 20 m. Mix in basal fertilizer (page 22). Plastic mulch is optional.
3. **Soak seeds** in clean water overnight. Sow right away, or air-dry first.
4. **Sow** 2–3 seeds per hill, 1 cm deep, 20 cm between hills and 25 cm between rows. Replant missing hills right away. Thin to 2 plants per hill 15 days after planting.
5. **Water** whenever the soil is dry or plants wilt.
6. **Hill up** one month after the plants come up, covering the base of the plants. Hand-pull the weeds between plants.
7. **Harvest** young, tender pods 7–10 cm (3–4 inches) long, 4–6 days after the flower opens. Pick every 2–3 days; old pods get tough and slow down the plant.

Pests and diseases

Pests: cotton stainer and stink bugs. Diseases: cercospora leaf blight, powdery mildew, fruit rot and root-knot nematode. Uproot and bury sick plants, keep the field clean, and rotate. Root-knot nematodes build up when okra follows okra.

Source: DA RFU VII Guide on Vegetable Production / ATI-VII Okra Production Guide (2020); BPI Okra Production Guide; DA RFO 02 Okra Production Guide.

Okra: reading your numbers

Find each of your averaged numbers in its row.

	✕ LOW	! SLIGHTLY LOW	✓ GOOD	▲ HIGH
pH	below 5.0 too sour	5.0 – 5.4 a bit sour	5.5 – 7.0 best 6.0–6.8	above 7.0 alkaline
N mg/kg	below 25	25 – 39	40 – 90	above 90
P mg/kg	below 12	12 – 19	20 – 50	above 50
K mg/kg	below 60	60 – 99	100 – 220	above 220

Example

Average of 5 readings: **pH 6.2 · N 32 · P 14 · K 150**

- pH 6.2 and K 150 → ✓ GOOD
- N 32 → ! SLIGHTLY LOW
- P 14 → ! SLIGHTLY LOW

Two nutrients are short, so the code is **3120** (several low). Add the extra 14-14-14 at basal and test again 2 weeks after the seedlings come up.

Why these numbers?

pH 5.5–7.0 from the DA RFU VII/ATI-VII and BPI okra guides. Okra flowers and fruits for many weeks, so it needs steady nitrogen and good phosphorus for pods. Since phosphorus is the most common shortage in Leyte soils (page 6), okra growers should watch the P row closely.

Okra: standard fertilizer plan

Use this plan when all four readings are ✓ GOOD.

WHEN	PER BED (1 × 20 M)	PER HECTARE
Basal at bed making	1 sack dried chicken manure **400 g complete 14-14-14** (about 1½ cups) Mix into the soil or cover with soil before sowing.	4 bags 14-14-14 + manure
Side-dress 1 about 30 days, first flower buds	**100 g urea** (½ cup) in a shallow furrow 5–10 cm from the plants. Cover with soil and water. Do this when you hill up.	1 bag urea
Side-dress 2 about 55–60 days, during harvest	**100 g urea** + **50 g muriate of potash 0-0-60**, same way.	1 bag urea ½ bag 0-0-60

Tips

- If the harvest goes on longer than 2 months, give another 100 g urea every 3–4 weeks while the plants are still flowering.
- Instead of dry side-dressing, you may dissolve the fertilizer in 16 L water and pour about 150 mL beside each hill (DA drench method).
- Yellowing older leaves during harvest usually means the plant needs nitrogen. Test to be sure.

Basal method, drench and side-dress timing: DA RFU VII / ATI-VII and BPI okra guides. Basal 2–3 bags 14-14-14 per ha plus organic fertilizer: DA RFO 02 Okra Production Guide. Plan supplies about 74-28-43 kg N-P_2O_5-K_2O per ha from fertilizer, plus manure.

Okra: Fix-it codes

Amounts are **in addition to** the standard plan. For ! SLIGHTLY LOW use **half**.

3101 All good	Follow the standard plan (page 22). Test again 2 weeks after side-dress 1.
3102 Too sour	**Lime first.** 4 kg agricultural lime or dolomite per bed (2 kg if only slightly sour), mixed into the top 15 cm and watered. Wait 2 weeks before sowing. See page 24.
3103 Alkaline	No lime or wood ash. Use **ammonium sulfate 21-0-0** instead of urea (2 g for every 1 g urea). Add 1 extra sack of compost.
3110 N low	Add **100 g urea** per bed, in two parts one week apart (or with each side-dress), covered and watered in.
3111 P low	Mix **300 g ammonium phosphate 16-20-0** (or 400 g Solophos 0-18-0) into the rows before sowing. On a growing crop, place it in a furrow beside the row, cover and water.
3112 K low	Add **100 g muriate of potash 0-0-60** per bed at basal, or split between the two side-dresses.
3120 Several low	Add **300 g more 14-14-14** and 1 more sack of manure or compost at basal. Test again 2 weeks after the seedlings come up and fix what is still low. If pH is also low, do 3102 first.
HIGH ▲ any	N high: skip the next urea side-dress. P high: use urea + 0-0-60 instead of 14-14-14. K high: skip 0-0-60.

Fixing sour (acidic) and alkaline soil

Lime is not fertilizer. It does not feed the plant; it **unlocks** the food already in the soil and the fertilizer you add later. Many Leyte soils are sour because of heavy rain and years of chemical fertilizer.

How much lime?

YOUR PH READING	PER BED (20 M²)	PER HECTARE
! SLIGHTLY LOW (a bit sour)	**2 kg** (about 7 cups)	1 ton
✕ LOW (too sour)	**4 kg** (about 13 cups)	2 tons
Too sour, heavy clay soil	up to 6 kg	up to 3 tons

Sandy soils need less lime than clay soils. These are starting amounts that raise pH by roughly half to one unit. A DA soil lab test gives the exact lime requirement for your field.

How to lime

1. Use **agricultural lime** (ground limestone) or **dolomite**. Dolomite also adds magnesium.
2. Spread it evenly over the bed **2–4 weeks before planting**.
3. Mix it into the top 15 cm of soil and water it in. Lime only works when it is mixed with moist soil.
4. **Wait at least 2 weeks** before applying urea or ammonium fertilizer. Lime and urea together lose nitrogen to the air.
5. Test pH again after 3–4 weeks. Do not lime again within the same season unless the reading is still low.

Alkaline soil (pH too high)

- Do not add lime, dolomite or wood ash.
- Work in compost or well-rotted manure every season.
- Use **ammonium sulfate 21-0-0** as your nitrogen source; it slowly lowers pH. Use 2 g of ammonium sulfate for every 1 g urea in the plan.

Fertilizers and how to measure them

PRODUCT	N-P-K	USE IT FOR
Complete	14-14-14	Basal feeding of all crops; "several low"
Urea	46-0-0	Nitrogen low; side-dressing leafy crops
Ammonium sulfate	21-0-0	Nitrogen on alkaline soil
Ammonium phosphate	16-20-0	Phosphorus low (also gives some N)
Solophos	0-18-0	Phosphorus low
Muriate of potash (MOP)	0-0-60	Potassium low; kamote roots
Agricultural lime / dolomite	—	Sour soil (raises pH)

The three numbers are the percent of N, P_2O_5 and K_2O in the bag. One 50-kg bag of 14-14-14 has 7 kg of each.

No scale? Measuring by spoon and cup

PRODUCT	1 LEVEL TABLESPOON	1 LEVEL CUP (250 ML)
Urea 46-0-0	≈ 10 g	≈ 190 g
14-14-14, 16-20-0, 21-0-0	≈ 15 g	≈ 260 g
Muriate of potash 0-0-60	≈ 15 g	≈ 260 g
Lime / dolomite powder	≈ 12 g	≈ 300 g

Granule sizes differ between brands. Best practice: weigh one cup of your fertilizer on a store scale once, and mark the level on your scoop.

Quick conversions used in this book

- 1 bed = 1 m × 20 m = 20 m² = 1/500 of a hectare
- 1 bag (50 kg) per hectare = **100 g per bed**
- 100 g urea ≈ ½ cup · 400 g 14-14-14 ≈ 1½ cups · 50 g 0-0-60 ≈ 3 tbsp

Safety: wear gloves; wash hands before eating; keep fertilizer away from children, animals and drinking water; never store it with food or seeds.

Compost and the DA soil laboratory

Feed the soil, not only the plant

Soil tests in Leyte and Southern Leyte keep finding **low organic matter** and sour soil, linked to years of chemical-only fertilizing. Municipal agriculture officers in Region VIII recommend locally available organic amendments such as **poultry manure and rice hull**.

- Add **1–2 sacks of compost or manure per bed every season**. It holds water and fertilizer so less is washed away by the rain.
- Rice-hull ash and carbonized rice hull loosen clay and add some potassium.
- Use well-rotted manure only. Fresh manure burns roots.
- Mulch with rice straw to protect the soil from heavy rain.

Getting a laboratory soil test

The SakaSense probe is for quick checks. A lab test is more complete (it includes organic matter and the exact lime requirement). Get one **for every new field and once a year**.

1. Take soil from **10–15 spots** in a zig-zag across the field, 0–20 cm deep, using a clean spade.
2. Mix it all in a clean pail. Take about **1 kg** of the mixed soil.
3. Dry it in the shade (not in the sun) and remove stones and roots.
4. Put it in a clean plastic bag and label it: your name, barangay, field, crop planned, date.
5. Bring it to the **DA RFO 8 Regional Soils Laboratory**, or ask your City/Municipal Agriculture Office about the DA **Mobile Soil Laboratory** schedule.

Reading a BSWM lab report

RATING	P BRAY 1 (PPM)	P OLSEN (PPM)	K (PPM, COLD H_2SO_4)
Low	0 – 2	0 – 6	0 – 35
Moderately low	2.1 – 6	6.1 – 10	36 – 55
Moderately high	6.1 – 10	10.1 – 15	56 – 75
High	10.1 – 15	15.1 – 20	above 75

Lab methods are different from the probe, so do not compare the numbers one-to-one. Compare the **rating** (low / high). Ratings from the BSWM Leyte nutrient status maps (2018).

Code finder

PROBLEM	KANGKONG	PECHAY	KAMOTE	OKRA
All good	7201	7101	2301	3101
Too sour (acidic)	7202	7102	2302	3102
Alkaline	7203	7103	2303	3103
Nitrogen low	7210	7110	2310	3110
Phosphorus low	7211	7111	2311	3111
Potassium low	7212	7112	2312	3112
Several low	7220	7120	2320	3120
Fix-it page	11	15	19	23

How a code is chosen: **pH comes first.** If pH is low or slightly low, the code is "too sour" (for sweet potato, only when pH is × LOW, below 4.5). If pH is fine and two or more of N, P, K are low or slightly low, the code is "several low". Otherwise the code names the one nutrient that is short.

My soil record

DATE	BED / CROP	PH	N	P	K	CODE	WHAT I DID

Sources and contacts

Region VIII soil data

- Bureau of Soils and Water Management (BSWM). Province of Leyte: Soil pH Map; Nutrient Status Maps, Phosphorus and Potassium (Key Corn Areas), 1:130,000. Soil Health Assessment – National Soil Sampling and Testing Project Phase III, with DA-RFO 8 and LGUs. 2018.
- DA Regional Field Office 8 Mobile Soil Laboratory results, Southern Leyte (Daily Tribune, 29 May 2026).
- Visayas State University research on Leyte soils: degraded upland soils are acidic (pH near 5) with low organic matter, N and P.

Crop practice and fertilizer rates

- DA Regional Field Unit VII, Guide on Vegetable Production; ATI-VII Kangkong, Pechay and Okra Production Guides (2020).
- Bureau of Plant Industry, Kangkong and Okra Production Guides. DA RFO 02 Okra and Sweet Potato Production Guides. DA Caraga pechay techno guide.
- Asio L.G., Tamayo N.V., dela Cruz N.E. Response of a traditional sweetpotato variety to fertilization in Leyte, Philippines. Tropentag 2018 (VSU / CLSU).
- Optimum fertilizer rate for kangkong on acid Ultisols (SANREM CRSP / IPB). Philippine pechay fertilizer study using 60-40-60 kg/ha as recommended rate.

Where to get help

- **Your City / Municipal Agriculture Office**: agricultural technicians, seeds, Mobile Soil Laboratory schedule
- **DA Regional Field Office 8**, Tacloban City: Regional Soils Laboratory
- **VSU PhilRootcrops**, Baybay City, Leyte: clean sweet potato planting material and root crop advice
- **SakaSense Venture Solutions**, Tacloban City: device and Kasaka app help · sakasense.com

The reading boxes for N, P and K are set for the SakaSense probe, using the BSWM ratings and the research above. They are a guide, not a guarantee. Check them against a DA laboratory test for your field, and ask your agriculture technician when unsure.

SakaSense Venture Solutions · Tacloban City, Leyte
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